How Reliable Are Next-Day Streamflow Forecasts During High Flows on the Potomac River?

Introduction. River managers need to know both how much water to expect tomorrow and how much confidence to place in that forecast when flows rise. We examined whether the accuracy of next-day forecasts across all river conditions also held on days of high discharge on the Potomac River near Washington, DC.

Methods. We used daily mean discharge from US Geological Survey gauge 01646500 from 1990 to 2025. Four neural networks used the preceding 30 days of discharge and seasonal information to predict the next day's flow. We trained the models on 1990–2009 data, validated them on 2010–2014 data, and calibrated a Bayesian ensemble on 2015–2019 data. We evaluated forecasts on 2,192 days in 2020–2025 against a baseline that carried the previous day's flow forward. We defined high flow as discharge above 797.4 m^3/s, the 90th percentile of the training observations.

Results. The stacked long short-term memory network achieved the lowest root mean square error, reducing it by 23.7% relative to the baseline, from 145.8 to 111.2 m^3/s. It anticipated 11 of 28 crossings into high flow one day ahead, missed 17, and produced four false alarms. The ensemble's 90% prediction interval contained the observed discharge on 93.0% of all test days, but on only 73 of 101 observed high-flow days (72.3%).

Conclusions. At this gauge, lower overall forecast error coexisted with frequent missed transitions into high flow. Reporting missed crossings, false alarms, and interval coverage during high flows gives water managers a clearer account of forecast performance than an average error alone.

Keywords Streamflow forecasting, Potomac River, high flow, machine learning, prediction intervals